

Matrix Notes

COSC2753

| SEMESTER 1 2025 | MACHINE LEARNING

Vector Operations

Summation (vectors must be same dimensions)

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} a + d \\ b + e \\ c + f \end{bmatrix}$$

Transpose (“flipping”)

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}^T = [a_1 \ a_2 \ a_3], \quad [b_1 \ b_2 \ b_3]^T = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Matrix Operations

Summation

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{bmatrix} + \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix} = \begin{bmatrix} a_{1,1} + b_{1,1} & a_{1,2} + b_{1,2} \\ a_{2,1} + b_{2,1} & a_{2,2} + b_{2,2} \\ a_{3,1} + b_{3,1} & a_{3,2} + b_{3,2} \end{bmatrix}$$

Transpose

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{bmatrix}^T = \begin{bmatrix} a_{1,1} & a_{2,1} & a_{3,1} \\ a_{1,2} & a_{2,2} & a_{3,2} \end{bmatrix}$$

Matrix Multiplication

“Rows by columns”

A has dimensions n by D, and **B** has D by k

$$\mathbf{AB} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,D} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,D} \end{bmatrix} \begin{bmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,k} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{D,1} & b_{D,2} & \cdots & b_{D,k} \end{bmatrix} = \mathbf{C}$$
$$\mathbf{C} = \begin{bmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,k} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,1} & c_{n,2} & \cdots & c_{n,k} \end{bmatrix} \text{ where } c_{i,j} = \sum_{d=1}^D a_{i,d} b_{d,j}$$

Not all multiplication possible

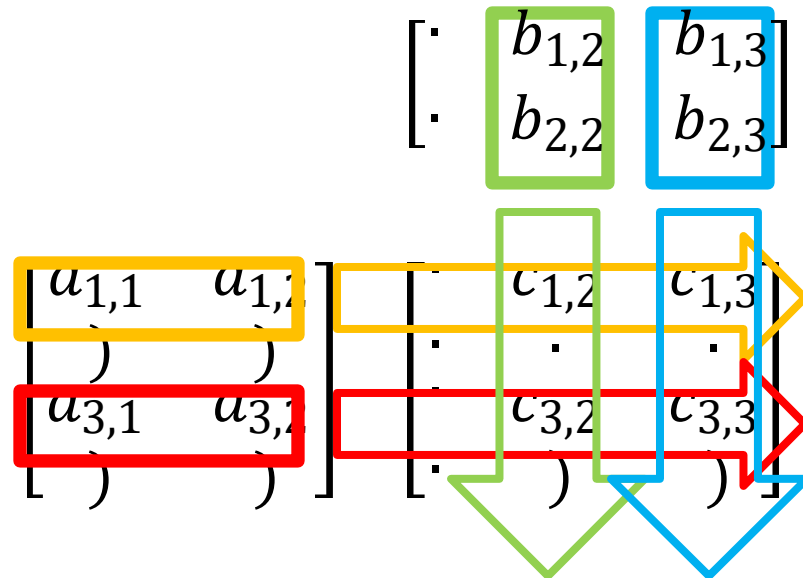
Matrix Multiplication

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ \vdots & \vdots \\ a_{3,1} & a_{3,2} \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} \cdot & b_{1,2} & b_{1,3} \\ \cdot & b_{2,2} & b_{2,3} \end{bmatrix} = \begin{bmatrix} \cdot & c_{1,2} & c_{1,3} \\ \vdots & \vdots & \vdots \\ \cdot & c_{3,2} & c_{3,3} \\ \vdots & \vdots & \vdots \end{bmatrix}$$

4x4 matrix

2x4 matrix

4x2 matrix



$$c_{1,2} = a_{1,1}b_{1,2} + a_{1,2}b_{2,2}$$

$$c_{3,3} = a_{3,1}b_{1,3} + a_{3,2}b_{2,3}$$

$$c_{1,3} =$$

$$c_{3,2} =$$